

Maurice Toney Jr.

Software Engineer | Python, Java, C++, JS

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SUMMARY

Software Engineer with experience building full-stack applications using Python, Java, and JavaScript. I bring the ability to deliver scalable solutions with modern frameworks such as Spring Boot and React, along with hands-on experience in machine learning and cloud deployment on Azure.

PROJECTS

AI Podcast Clipper

- Enhanced a cloud-based AI video-clipping platform built with Next.js and Python, implementing new functionality across both the frontend and GPU-powered backend.
- Implemented asynchronous media processing workflows using Inngest, enabling reliable orchestration of long-running transcription and video processing tasks.
- Extended the media ingestion pipeline by adding YouTube video support, integrating download, validation, and cloud storage into the existing processing workflow.
- Integrated AWS S3 and Supabase/PostgreSQL to support persistent storage, job tracking, and retrieval of generated media assets.
- Configured and deployed GPU-accelerated processing services on Modal and the frontend on Vercel, resolving dependency, deployment, and infrastructure issues to achieve a production-ready deployment.
- Diagnosed and resolved integration issues across a distributed application stack, including cloud services, environment configuration, API communication, and third-party dependencies.

Delivery Routing Program

- Used Python to build a delivery route optimization system that calculates efficient paths for real delivery addresses using constraint-based algorithms.
- Implemented Dijkstra's algorithm with custom heuristics to score and select optimal routes based on distance, traffic, and delivery windows.
- Architected event-driven simulation engine with CLI interface for route analysis and performance reporting.
- Used domain-driven design to separate external ingestion of spreadsheet data from the domain-specific logic relating to package delivery.
- Supported future refactoring concerns through unit tests built with pytest

Hotel Reservation Web App

- Enhanced a full-stack Spring Boot and Angular production hotel reservation system, fixing critical API bugs and rebuilding UI components for a better user experience
- Optimized API response times and implemented CORS security for safe cross-origin data transmission
- Added internationalization support through resource bundling for multi-language deployment
- Containerized the application with Docker and deployed to the cloud using Azure with an automated CI/CD pipeline
- Used Postman and functional testing to confirm the synchronization of the frontend and backend programs.

Grade Predictor AI Model

- Trained a regression-based machine learning model that predicts student success using Scikit-learn, XGBoost, Numpy, and Pandas
- Processed a 20+ parameter dataset found on Kaggle and performed feature selection and tuning to make the data informative
- Used Jupyter Notebook, Matplotlib, and Seaborn to visualize data features, model performance, and other relevant characteristics surrounding both the model and data
- Designed the program to support multiple machine learning algorithms and display performance in the console and logs

EDUCATION

Bachelor of Science in Computer Science

Western Governors University | Aug 2025

SKILLS

Programming Languages: Python, Java, C++, JavaScript, Bash, SQL

Frameworks & Libraries: Spring Boot, Django, Node.js, React, Next.js, Angular, PyTorch, TensorFlow, scikit-learn, Pandas

Databases & Storage: PostgreSQL, MySQL, Redis, Prisma, AWS S3

Cloud, DevOps & Infrastructure: Azure, Docker, Kubernetes, Git, Linux, CI/CD, Vercel, Inngest

Software Engineering: Object-Oriented Programming, Domain-Driven Design, Software Design, Distributed Systems, REST API Development, Data Structures & Algorithms, Web Development

Data & AI: Data Engineering, Machine Learning, Power BI, Prompt Engineering

Tools & Methodologies: Agile Development, Jira

PROFESSIONAL EXPERIENCE

Insurance Claims Adjuster

Pilot Catastrophe Services | Jan 2021 – Present

- Process 8 - 20 insurance claims daily while maintaining compliance with state regulations and company policies.
- Investigate and resolve claims by analyzing policy documentation, gathering evidence, and applying coverage determination logic.
- Thoroughly document decisions and pertinent facts surrounding the resolution.
- Prioritize concurrent claims processing based on severity levels and business impact assessments.
- Employ time-management techniques to operate in high-volume periods, such as during catastrophic events.